Chapter 8: Analysis of Variance (ANOVA)

Statistics for Environmental Science

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Chapter Summary **22**

ANOVA is analysis of variance

ANOVA stands for analysis of variance. It tests whether differences among several group means are larger than we'd expect from chance alone. Chapters 6 and 7 compared two groups. This chapter builds the one-way ANOVA by hand, runs the full workflow on one example through post-hoc comparisons and a rank-based alternative, and ends with repeated-measures ANOVA.

With two groups, a t-test already answers that question. ANOVA extends the same logic to three or more groups. Environmental studies often compare that many, such as several sites, treatments, or time periods.

> **Note: Where ANOVA came from**
>
> Sir Ronald Fisher invented the ANOVA (Salsburg 2001) and wanted to publish it in the journal Biometrika. The editor at the time was Karl Pearson, of Pearson's r, and the two men disliked each other, so Pearson refused to publish the new test. Fisher published it in the Journal of Agricultural Science instead. The feud carried into the next generation. Years later, Karl Pearson's son Egon Pearson, working with Jerzy Neyman, revamped Fisher's ideas and re-cast them into what we now call null vs. alternative hypothesis testing. Fisher objected to that too.

Why not run multiple *t*-tests?

Suppose we're comparing classroom CO_2 concentrations (ppm) under three ventilation settings: windows open, mechanical ventilation, and no ventilation change. With three groups, there are three possible pairwise comparisons: windows open vs. mechanical ventilation, windows open vs. no change, and mechanical ventilation vs. no change.

We could run a separate t-test for each. The problem is that every test carries its own 5% chance of a false positive (a Type I error, Chapter 5), and those chances accumulate across tests. This follows from the independence multiplication rule in Chapter 3: if we run m independent tests at $\alpha = 0.05$, the probability of at least one false positive across all of them is

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^m.$$

With three groups, that's three tests. The **familywise error rate** is the chance of at least one false positive across a set of tests. For three tests it is 14%, compared with 5% for a single test:

$$1 - (1 - 0.05)^3 = 0.14.$$

Number of groups (k)	Pairwise tests (m)	Familywise error rate
3	3	$1 - 0.95^3 = 0.14$
4	6	$1 - 0.95^6 = 0.26$
5	10	$1 - 0.95^{10} = 0.40$

Even with only five groups, there's a 40% chance of turning up at least one "significant" difference somewhere purely by chance, even if every population mean is identical. Figure 1 shows how the chance of at least one false positive grows as the number of independent t-tests grows.

ANOVA solves this by replacing many separate t-tests with *one* test that controls the overall error rate. It asks a single question:

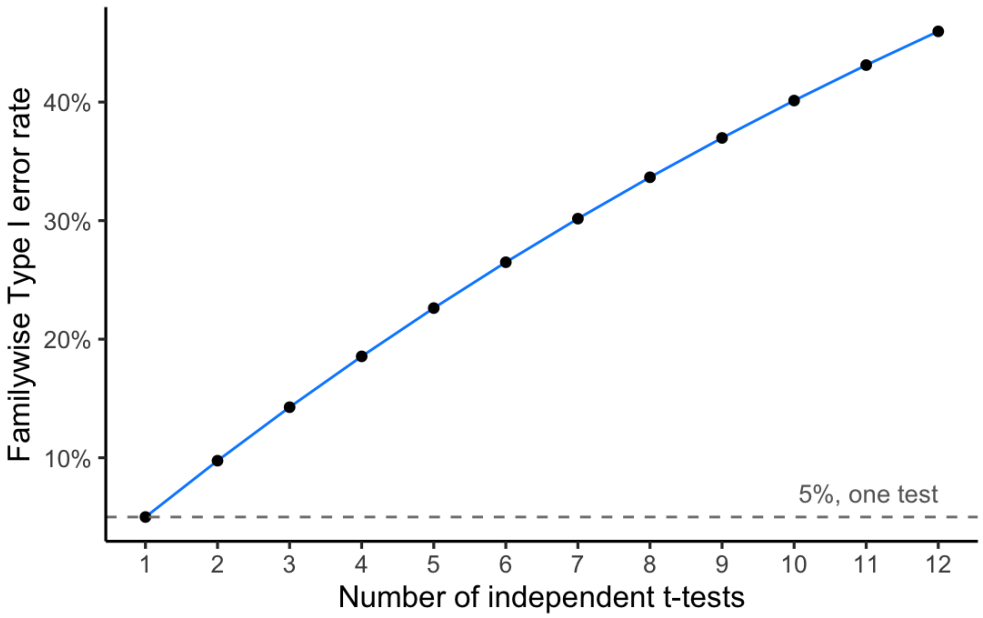


Figure 1: Familywise Type I error rate as the number of independent t-tests increases, at $\alpha = 0.05$ per test.

$$H_0 : \mu_1 = \mu_2 = \mu_3 \qquad H_a : \text{at least one mean differs}$$

Just like the t-test, different research designs call for different kinds of ANOVA: a one-way (between-subjects) ANOVA when every unit is measured once, and a repeated-measures ANOVA when the same units are measured under multiple conditions. This chapter covers the one-way case and ends with a short look at repeated measures. Chapter 10 extends the one-way case to two factors.

One-way ANOVA

The one-way ANOVA (also called a between-subjects ANOVA) is the simplest version of the test. In ANOVA an independent variable is called a **factor**, and the groups it defines are its **levels**. A one-way ANOVA has one factor with two or more levels, and every unit is measured once. With two levels it gives the same result as the pooled t-test, so in practice it is used for three or more.

Table 1 sets the one-way ANOVA beside the two other kinds of ANOVA this book covers.

Table 1: Three kinds of ANOVA.

ANOVA type	Factors and units	Design
One-way ANOVA	One factor with two or more levels, with different units in each group	Between-subjects
Repeated-measures ANOVA	One factor with two or more levels, with the *same* units measured at every level	Within-subjects

ANOVA type	Factors and units	Design
Two-way (factorial) ANOVA	Two factors, each with two or more levels, fully crossed (Chapter 10)	Factorial

ANOVA's test statistic is F, named for Fisher, who invented ANOVA. Like t, F compares the size of an effect to the amount of variation you'd expect from sampling error alone. With exactly two groups, a t-test and a one-way ANOVA give the same answer, because $t^2 = F$.

$$F = \frac{\text{measure of effect}}{\text{measure of error}}$$

The t-test puts a difference between two means over its standard error. F uses variances for both the numerator and the denominator. It divides the variance you can explain (differences between group means) by the variance you can't (leftover noise within each group).

$$F = \frac{\text{Can explain}}{\text{Can't explain}}$$

When the two are about the same size, $F \approx 1$, and group differences are no bigger than ordinary sampling noise. When explained variance is much larger than unexplained variance, F climbs well above 1, which is evidence that the groups differ. When it's smaller, F drops below 1. An F of 5 means the group means vary five times as much as within-group noise alone would make them vary. Both parts of F are variances, so "five times as much" compares squared distances, not raw distances. An F of 0.6 means they vary less than that noise alone would typically produce.

Figure 2 shows both situations with the same noise. On the left, the group means differ by no more than the scatter within each group would produce. On the right, the means are separated by much more than that scatter. F measures that separation relative to the within-group scatter.

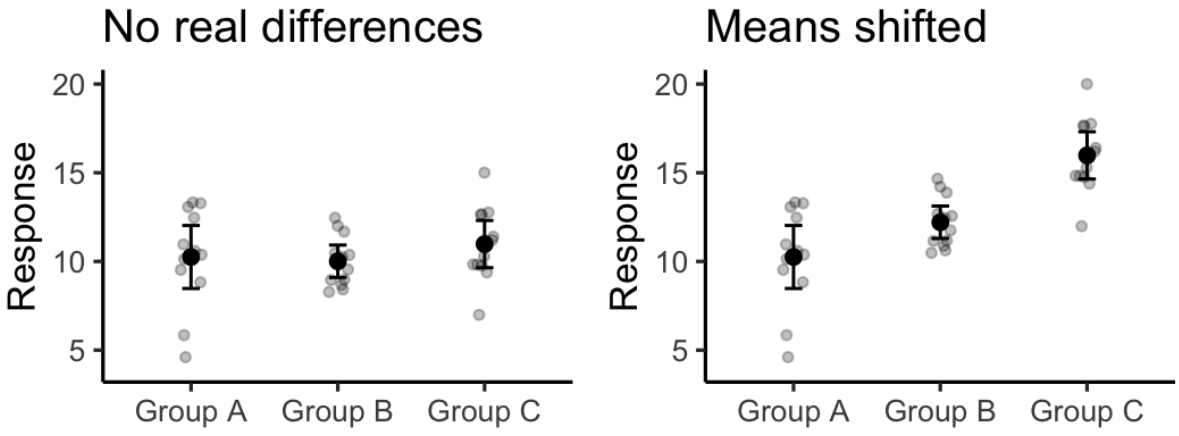


Figure 2: The same within-group noise with no real differences (left) and with the group means shifted (right). Three groups of 12; small points are individual scores, large dots are group means, bars are 95% confidence intervals.

Computing the *F*-value

ANOVA splits the total variation in the data into two sources: variation between groups, which group membership can explain, and variation within groups, which it can't. If we can measure those two parts separately, we can turn them into a ratio and get an F-value.

$$\text{Total variation} = \text{Variation between groups} + \text{Variation within groups}$$

To make that precise we need an actual measure of variation. As in Chapter 2, that's the sums of squares:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

The next sections compute each piece for a small example with nine scores. We find SS_{Total} and SS_{Between}, get SS_{Within} by subtraction, and check it directly. Then we divide each sum of squares by its degrees of freedom and take the ratio to get F.

SS Total

SS_{Total} is the sum of the squared differences between each score and the grand mean, the mean of all the scores. It captures all of the variation in a dataset. Writing Y_{ij} for score i in group j and $\bar{Y}_{\text{grand}}$ for the grand mean,

$$SS_{\text{Total}} = \sum (Y_{ij} - \bar{Y}_{\text{grand}})^2.$$

Suppose we have three groups, A, B, and C, with 3 scores in each:

Table 2: Deviations from the grand mean.

Group	Score	Deviation from grand mean	Squared deviation
A	20	13	169.00
A	11	4	16.00
A	2	-5	25.00
B	6	-1	1.00
B	2	-5	25.00
B	7	0	0.00
C	2	-5	25.00
C	11	4	16.00
C	2	-5	25.00
Sums	63	0	302.00
Means	7	0	33.56

The data are in long format, which means each row is a single score. The mean of all of the scores is the **grand mean**, calculated in the table as 7.

The deviation column holds every score's distance from the grand mean, and the last column squares those distances. Squaring matters because the raw deviations always sum to zero (the mean is the balancing point of the data), so their sum can't measure spread. Squaring turns every deviation positive and, as a side effect, weights larger deviations more heavily.

Adding up all of the squared deviations gives the sum of squares (SS) from Chapter 2. Here $SS_{\text{Total}} = 302$, the first piece of the partition:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}, \qquad 302 = SS_{\text{Between}} + SS_{\text{Within}}.$$

We only need to compute one of the two remaining pieces directly. The other follows by subtraction.

SS Between

SS_{Total} captured all of the variation in the data. SS_{Between} is the part of the total variation that group membership can explain. It measures how far each group's mean is from the grand mean.

To find it, replace every score with its own group's mean, then square and sum each group mean's distance from the grand mean. Writing $\bar{Y}_j$ for the mean of group j and n_j for its number of scores,

$$SS_{\text{Between}} = \sum_j n_j(\bar{Y}_j - \bar{Y}_{\text{grand}})^2.$$

The n_j is there because every score in group j is replaced by the same group mean, so each group's squared distance counts once per score. Here every group has 3 scores. With unequal groups, larger groups count more.

Table 3: Group means replace the scores.

Group	Score	Group mean	Group mean minus grand mean	Squared
A	20	11	4	16
A	11	11	4	16
A	2	11	4	16
B	6	5	-2	4
B	2	5	-2	4
B	7	5	-2	4
C	2	5	-2	4
C	11	5	-2	4
C	2	5	-2	4
Sums	63	63	0	72

The group mean column replaces each score with its group mean. Group A's mean is 11, so there are three 11s, one per observation in group A. Groups B and C both average to 5, so the rest of the column is 5s.

Within a group every value is now identical, so this column throws away the individual variation inside each group and keeps only the differences between groups, which is what SS_{Between} measures.

Finding the differences between each of those group-mean values and the grand mean, squaring them, and summing gives $SS_{\text{Between}} = 72$, the variation we can explain by group membership.

SS_{Within} is a sum of squares, so it cannot be negative. That means SS_{Between} can never be larger than SS_{Total}, and here 72 is smaller than 302.

SS Within

We already have $SS_{\text{Total}} = 302$ and $SS_{\text{Between}} = 72$, so we can solve for SS_{Within} by subtraction:

$$SS_{\text{Within}} = SS_{\text{Total}} - SS_{\text{Between}} = 302 - 72 = 230.$$

We can also calculate SS_{Within} directly from the data, both to see where it comes from and to check the subtraction.

$$SS_{\text{Within}} = \sum (Y_{ij} - \bar{Y}_j)^2$$

Table 4: Deviations from each group's own mean.

Group	Score	Group mean	Score minus group mean	Squared
A	20	11	9	81
A	11	11	0	0
A	2	11	-9	81
B	6	5	1	1
B	2	5	-3	9
B	7	5	2	4
C	2	5	-3	9
C	11	5	6	36
C	2	5	-3	9
Sums	63	63	0	230

This time, for each score we found the group mean first, then the leftover error. The fourth column is the difference between each score and its own group mean, and the last column squares it. Summing the last column gives SS_{Within}, the variation the group means can't explain.

SS_{Within} has several other names. It is also called the **error sum of squares**, since it is the variation left over after group membership has explained what it can. R labels its row `Residuals`. In the next sections we divide it by its degrees of freedom to get MS_{Within}, which some articles report as MSE (mean squared error).

Degrees of freedom

Degrees of freedom work the same way here as for the t-test. Every time we estimate something from the data, we use up a degree of freedom.

$df_{\text{Between}} = k - 1$, where k is the number of groups. With $k = 3$, $df_{\text{Between}} = 3 - 1 = 2$. Once we fix the grand mean, two of the three group means are free to be anything, but the third is determined by the other two.

$df_{\text{Within}} = N - k$, where N is the total number of scores. With $N = 9$ and $k = 3$, $df_{\text{Within}} = 9 - 3 = 6$. Estimating the 3 group means uses up 3 degrees of freedom, one per group. Within each group, the deviations from the group mean must sum to zero, so only 2 of a group's 3 deviations are free to vary. Three groups with 2 free deviations each give $df_{\text{Within}} = 6$.

$df_{\text{Total}} = N - 1$, and it splits the same way the sums of squares do: $df_{\text{Total}} = df_{\text{Between}} + df_{\text{Within}}$, here $8 = 2 + 6$.

Mean squares

We're building toward $F = \dfrac{\text{measure of effect}}{\text{measure of error}}$, and the first idea is to divide SS_{Between} by SS_{Within} directly. The problem is that each sum of squares grows with the number of pieces that go into it. SS_{Within} adds up 9 squared deviations and SS_{Between} is built from only 3 group means, so SS_{Within} tends to be larger because it has more terms. Dividing each one by its degrees of freedom turns it into an average, and two averages can be compared directly.

A **mean square** is a sum of squares divided by its degrees of freedom. It turns each sum of squares into an average. We divide by degrees of freedom rather than by n, because each group mean we estimate uses up a degree of freedom, as the previous section showed.

$$MS_{\text{Between}} = \frac{SS_{\text{Between}}}{df_{\text{Between}}} = \frac{72}{2} = 36$$

$$MS_{\text{Within}} = \frac{SS_{\text{Within}}}{df_{\text{Within}}} = \frac{230}{6} = 38.33$$

Both are variances. MS_{Within} measures how much individual scores vary around their own group's mean. MS_{Between} measures how much the group means vary, multiplied by the number of scores per group, which puts it on the same scale as MS_{Within}.

Calculate *F*

With MS_{Between} and MS_{Within} in hand, F is their ratio:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{36}{38.33} = 0.94$$

The ANOVA table

An **ANOVA table** lists the sums of squares, degrees of freedom, mean squares, and F for each source of variation. Table 5 gives the formulas, and the table below fills them in for the toy data.

Table 5: The one-way ANOVA table in formulas.

Source	df	SS	MS	F
Between groups	$k-1$	$\sum_j n_j(\bar{Y}_j - \bar{Y}_{\text{grand}})^2$	$SS_{\text{Between}}/df_{\text{Between}}$	$MS_{\text{Between}}/MS_{\text{Within}}$
Within groups	$N-k$	$\sum(Y_{ij} - \bar{Y}_j)^2$	$SS_{\text{Within}}/df_{\text{Within}}$	
Total	$N-1$	$\sum(Y_{ij} - \bar{Y}_{\text{grand}})^2$		

Table 6: R's ANOVA table for the toy data.

Source	df	SS	MS	F	p
groups	2	72	36.00	0.94	0.442
Residuals	6	230	38.33		

R labels the two rows `groups` (the between-groups row: what the means can explain) and `Residuals` (the within-groups row: what they can't). Other software uses slightly different labels for the same two rows. Apart from the p-value in the last column, the table contains only numbers we already computed. R leaves out the Total row. It organizes the numbers, and MS_{Between} (36) divided by MS_{Within} (38.33) gives the F-value. With $F = 0.94$, the group means vary no more than the noise within groups would make them vary, so these nine scores give no evidence that the groups differ.

What does *F* mean?

Every number in the ANOVA table so far was computed directly from the data except the p-value. It means the same thing as for every other test we've covered. The p-value is the probability of observing an F this large or larger if the null hypothesis (no real group differences) were true.

Since F is a sample statistic, we can find out how it behaves the same way we found out how t behaves, by simulating it many times under the null and looking at the resulting distribution.

To build intuition for the F distribution, consider a simulated monitoring study with three site types (A, B, C). Each site type has 10 sampling plots, for 30 total. We measure species abundance at each plot. All 30 plots are drawn from the same null population distribution (mean = 100, $\sigma = 10$), so site type has no real effect on abundance. Any differences between group means arise from sampling error alone.

We run this simulation 10,000 times. Each time we compute F from the sample data. The histogram of those 10,000 F-values, shown in Figure 3, is the null distribution of F for this design. The null population distribution gives the individual abundance scores (mean 100, $\sigma = 10$). The null distribution of F describes the F-values, one per experiment, that repeated experiments drawn from that population produce.

Before you look. Each bar in Figure 3 counts F-values from these null experiments. Decide whether most of them will fall near 0, near 1, or well above 1.

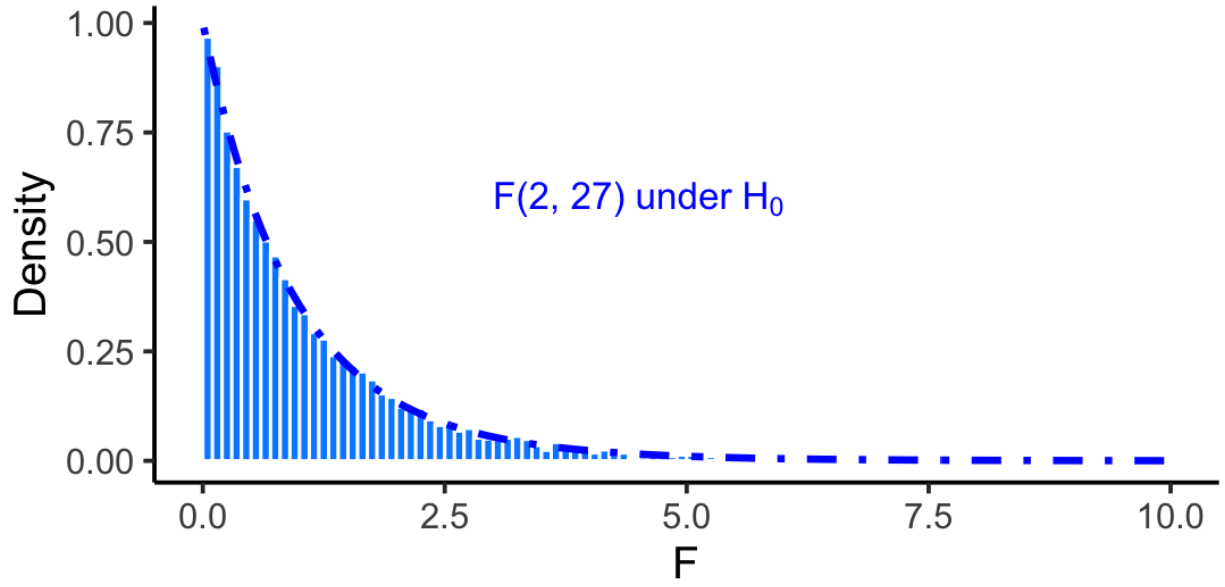


Figure 3: A simulation of 10,000 experiments from a null population distribution with no group differences. The histogram shows 10,000 F-values, one for each simulation. They show the values F takes when the null hypothesis is true. All of these F-values were produced by random sampling error. The blue dot-dash curve is the F distribution with 2 and 27 degrees of freedom.

The simulated null distribution of F has two properties. First, it never goes below 0, because F is a ratio of two variances and variances can't be negative. Second, it is right-skewed, and its exact shape depends on the numerator and denominator degrees of freedom.

Most simulated F-values here fall near or below 1, because under the null the two mean squares estimate the same thing. All 30 plots come from one null population distribution with $\sigma = 10$. MS_{Within} estimates σ^2 from the spread of scores inside each group. MS_{Between} estimates the same σ^2 from how far the group means stray from the grand mean, since with no real effect, sampling variation is the only thing moving them. Two estimates of one variance give a ratio near 1. When the population means really differ, MS_{Between} picks up those differences on top of σ^2, and F grows. Values of 5 or more occur in about 1% of the simulations.

From simulation to the F distribution. Everything in Figure 3 so far came from simulation. The **F distribution** is the theoretical distribution of F when the null hypothesis is true. Its shape is set by two degrees of freedom, the numerator df $(k-1)$ and the denominator df $(N-k)$. The blue dot-dash curve drawn over the histogram is the F distribution with 2 and 27 degrees of freedom ($k-1=2$ for the three site types, $N-k=27$ for the 30 plots), worked out from theory with no simulation at all. It follows the histogram closely, so from here on we use the curve. Software does the same. The p-value in an ANOVA table comes from the F distribution for the design's two dfs, and nothing is simulated.

Making decisions

We can use this null distribution of F to decide when a real experiment's result is too large to be explained by chance:

1. Set $\alpha = 0.05$.
2. Find the critical value of F for our design's numerator and denominator dfs, from the F distribution (in R, `qf(0.95, 2, 27)`).

Figure 4 marks that critical value on the histogram:

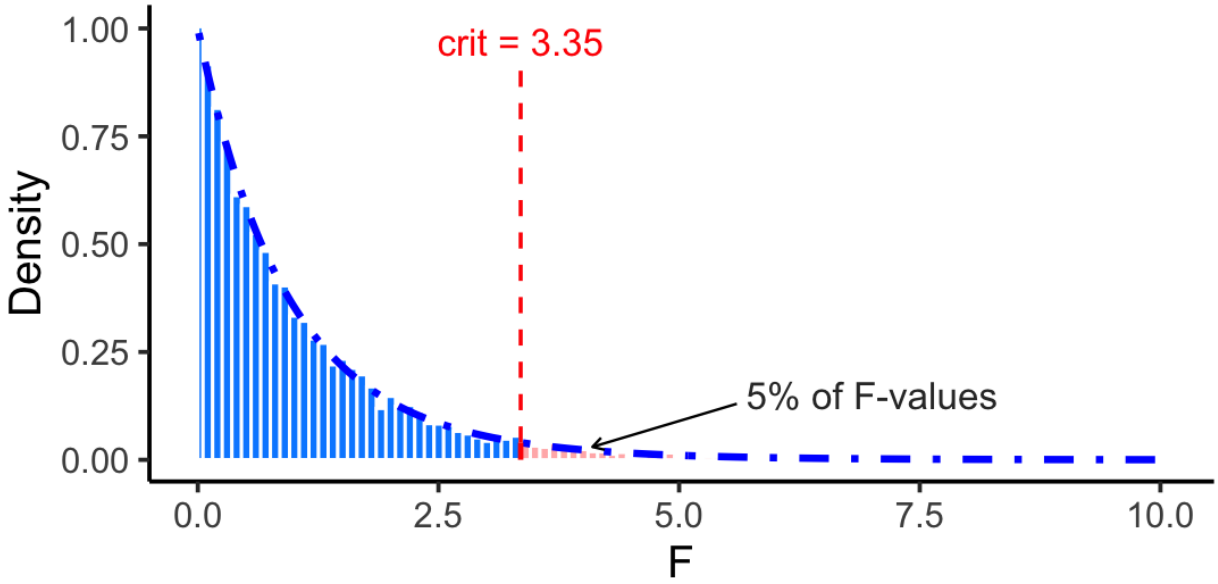


Figure 4: The critical value of F at $\alpha = 0.05$ for df = 2 and 27. Five percent of the F distribution lies beyond it, and about 5% of the simulated F-values do too.

Only 5% of the F distribution lies at or beyond 3.35. The F-test has one critical value, in the upper tail. Differences among the means in any direction make MS_{Between} larger, so they all push F up. A small F is consistent with H_0. The simulation agrees, with 4.8% of the 10,000 simulated F-values at or beyond it. That gives us a decision rule. If we ran the same design for real (30 plots, 10 per site type), we could treat $F \geq 3.35$ as evidence that the differences between site types are larger than chance would produce. An observed F of 3.4 or larger, for instance, would occur less than 5% of the time under the null, so seeing it would lead us to reject the null and conclude that site type mattered. The critical value and the p-value give the same decision. F is at or beyond the critical value exactly when its p-value is at or below α.

F and the means

An F-value and its p-value say whether the group means differ, but not what the means were. If a study reports $F(2, 27) = 6.0$, $p = 0.007$, we know only that an F this large would happen by chance less than 1% of the time. We still don't know which means differed, or by how much, without looking at the group means directly (the ANOVA table itself doesn't report them).

ANOVA is an omnibus test

An **omnibus test** is a single test that covers several comparisons at once, and ANOVA is one.

With three groups A, B, and C, there are three possible pairwise differences: A vs. B, B vs. C, and A vs. C. We could test each with a separate t-test, or ask the more general omnibus question with a single ANOVA:

Hypothesis of no differences anywhere: $\mu_A = \mu_B = \mu_C$

Any differences anywhere:

a. $\mu_A \neq \mu_B = \mu_C$ (A differs)
b. $\mu_B \neq \mu_A = \mu_C$ (B differs)
c. $\mu_C \neq \mu_A = \mu_B$ (C differs)
d. $\mu_A \neq \mu_B$, $\mu_B \neq \mu_C$, and $\mu_A \neq \mu_C$ (all three differ)

A small F with a large p-value means we don't reject the hypothesis of no differences. We have no evidence the three group means differ, and whatever gaps exist could easily be chance. A large F with a small p-value (below α) means we reject that hypothesis and conclude at least one group mean differs from the others. The omnibus test tells us *whether* the means differ, and a separate step tells us *which* ones.

What group means look like when *F* exceeds the critical value

Earlier we found a critical value of 3.35. Only 5% of F-values from this null distribution exceed it. The 10,000 simulated experiments kept only each F-value, not the group means behind it. To see those means, we reran the same null simulation and kept the first 10 experiments with $F > 3.35$, shown in Figure 5. Every one of them was still drawn from the same null population distribution, with no real group differences. Rejecting the null whenever F clears 3.35 is the right long-run decision rule, but every one of these particular simulations would be a Type I error if we did.

Every panel of Figure 5 now shows at least one group mean whose error bars don't overlap with another's, which looks like a convincing real difference. All ten are Type I errors. When this happens by chance, the data show a pattern, the F-value is large, and the p-value is small. A Type I error looks exactly like a real effect, and nothing in the output distinguishes them.

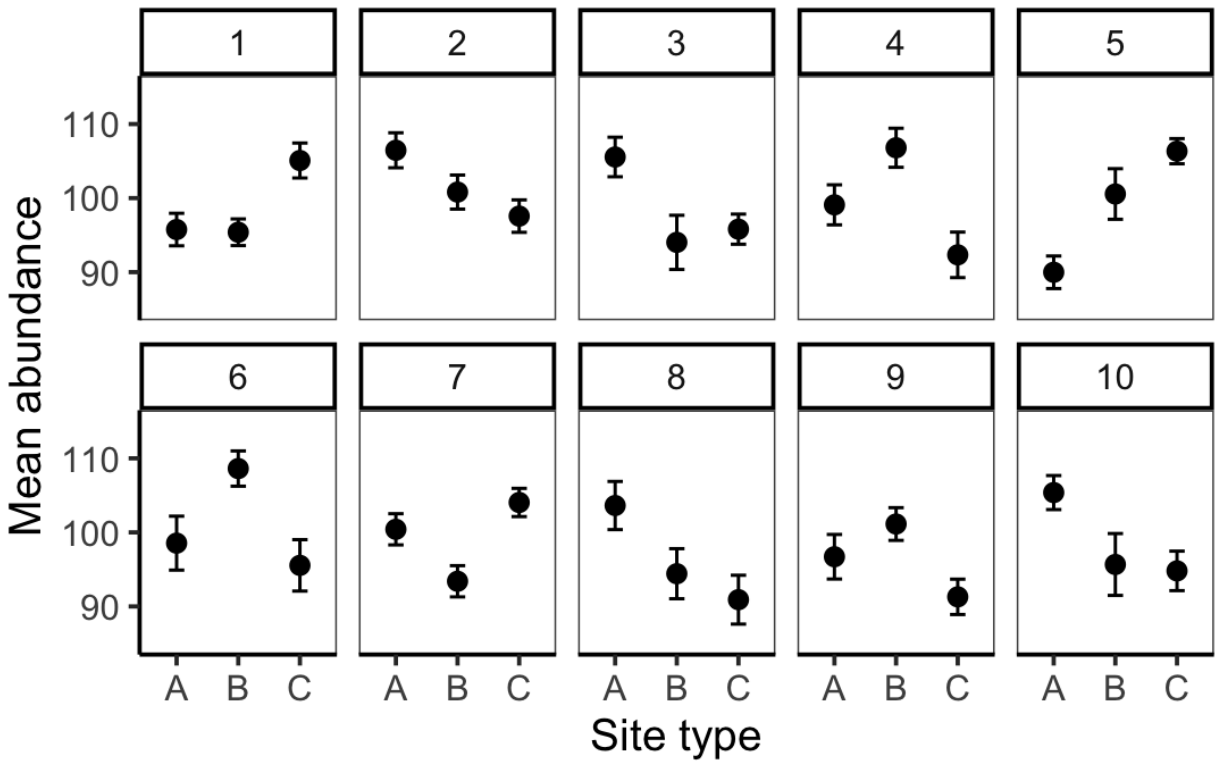


Figure 5: Group means from 10 null simulations in which F exceeded the critical value. Error bars show ±1 SE. Every one of them is a Type I error.

The full ANOVA workflow

In practice, software computes the ANOVA table. We computed one by hand so that each number in the table has a meaning when we read it.

The following example applies the full ANOVA workflow, from descriptive summaries through the omnibus F test to post-hoc comparisons, to a salamander study.

Salamander abundance across forest types

Plethodontid salamanders are sensitive indicators of forest condition, because they depend on moist microhabitats, abundant coarse woody debris, and stable soil temperatures, all features that decline with forest disturbance. A field ecologist surveys salamander abundance across three forest types, laid out in Figure 6. The counts in this example are invented. We built them so that old-growth plots average about 14 salamanders, secondary forest about 7, and managed plantation about 5, with 8 plots of each type.

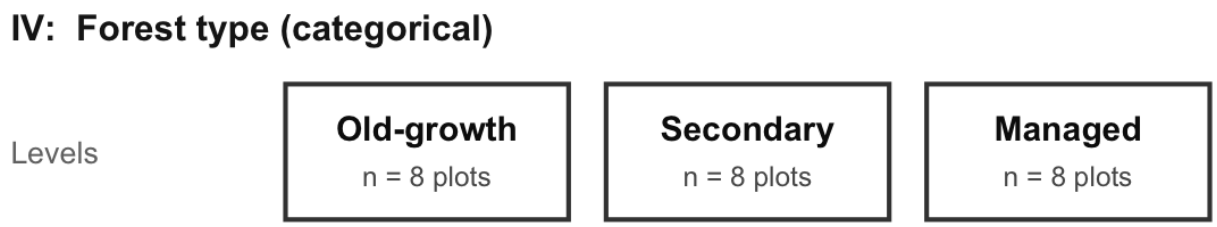


Figure 6: The salamander study as a design schematic. Structurally this is the two-group design from Chapters 4 and 6 with one more box added.

Compared with the two-group designs earlier in the book, only the number of groups has changed. ANOVA extends the t-test the same way, from two groups to three or more. More groups also means more pairwise comparisons, which is the multiple-comparisons problem from the start of the chapter.

The hypotheses are $H_0 : \mu_{\text{Old-growth}} = \mu_{\text{Secondary}} = \mu_{\text{Managed}}$ and H_a: at least one mean differs. Let's look at the data.

The bar graph shows a gradient. Old-growth plots have the highest mean abundance, managed plantations the lowest, and secondary forest is in between. The individual plot counts (dots) spread widely within each forest type. Old-growth plots range from 10 to 19 salamanders and secondary plots from 3 to 13, so even where a real difference exists, individual plots from different forest types overlap.

Checking ANOVA's assumptions

Before running the ANOVA, check whether its assumptions hold. Chapter 6 introduced independence as the assumption every test in this book shares, so we do not list it again here. ANOVA adds two assumptions of its own.

The first is **homogeneity of variance**, which means the groups have roughly equal spread around their own means. The second is approximately normal residuals. In ANOVA the model predicts each score with

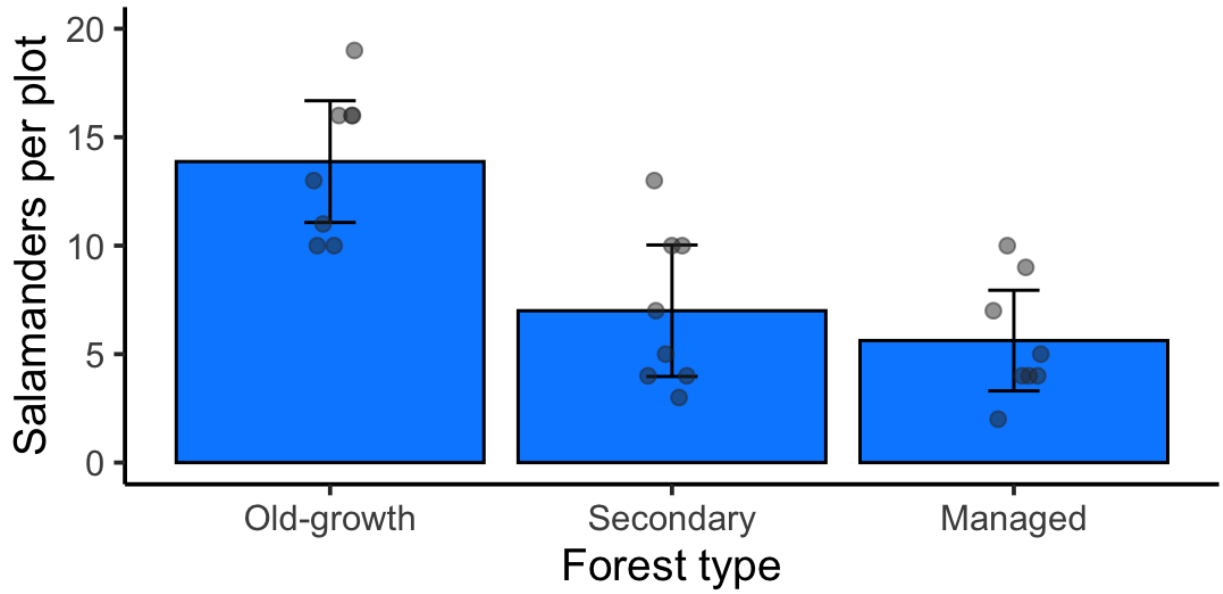


Figure 7: Mean salamander abundance (individuals per plot) across three forest types. Points show individual plot counts, and error bars show 95% confidence intervals.

its group mean, so a residual is the score minus its group mean, the same idea as the residuals in Chapter 2. Here each residual is a plot's count minus its forest type's mean, the same deviations that make up SS_{Within}, and the residuals should be roughly bell-shaped. Both matter because the F-test pools the within-group spread of every group into one error variance, MS_{Within}, and then compares F to a null distribution that assumes normal residuals. Pooling only makes sense when the groups have similar spread.

Levene's test checks homogeneity of variance. Its null hypothesis is that every group has the same variance, so a small p-value is evidence that the spreads differ. It reports its own F statistic, separate from the ANOVA's F. The Shapiro-Wilk test from Chapter 6 checks normality, here applied to the residuals.

```
car::leveneTest(abundance ~ forest_type, data = salamander_df)
```

Table 7: Levene's test for homogeneity of variance.

Statistic (F)	df	df residual	p
0.52	2	21	0.599

The residuals come from the fitted model, so we fit the ANOVA first and set its table aside until the checks are done.

```
aov_out <- aov(abundance ~ forest_type, data = salamander_df)
shapiro.test(residuals(aov_out))
```

Table 8: Shapiro-Wilk test on ANOVA residuals.

Statistic (W)	p
0.93	0.078

Levene's test (Levene 1960; Brown and Forsythe 1974) gives $F(2, 21) = 0.52$, $p = 0.599$, so there is no evidence of unequal variance across forest types. The Shapiro-Wilk test (Shapiro and Wilk 1965) on the model's residuals gives $W = 0.93$, $p = 0.078$, so there is no evidence against normality either. Both checks agree with the individual plots in Figure 7, which show similar spread across groups and no outliers far from the rest.

Formal assumption tests are evidence, not a gate. Chapter 6 made this point about Shapiro-Wilk, and it holds for Levene's test too. It's tempting to treat Levene's test as a strict gate, where significant means "stop" and not significant means "proceed." With a small sample (exactly when a violated assumption matters most), these tests have low power and will often say "no problem" even when there is one. With a very large sample (where ANOVA is fairly robust to mild violations anyway), they have high power and will flag trivial differences as "significant." Look at the data too, as in Figure 7, and check whether the groups have similar spread and whether any points are far from the rest.

In practice, one-way ANOVA is moderately robust to mild departures from these assumptions, especially when the design is **balanced**, with the same number of observations in every group, as here (8 plots per forest type). Be most cautious when small, unbalanced samples come with visibly skewed data or very different spreads. In that case, the nonparametric alternative below is the better choice.

Table 9 lists each assumption, how to check it, and what to use if it fails.

Table 9: ANOVA's assumptions, how to check them, and what to do if one fails.

Assumption	How to check	If it fails
Approximately normal residuals	Shapiro-Wilk test on the residuals, and a look at the scores in each group (Figure 7)	Kruskal-Wallis test
Homogeneity of variance	Compare the spread of each group in a plot, and Levene's test	Welch's ANOVA

With the assumptions reasonable, we run the one-way ANOVA to test whether these differences are larger than expected by chance alone.

```
summary(aov_out)
```

Table 10: One-way ANOVA table for salamander abundance by forest type.

Source	df	SS	MS	F	p
forest_type	2	312.58	156.29	14.60	< 0.001
Residuals	21	224.75	10.70		

With 24 plots the degrees of freedom are 2 and 21, and the critical value at $\alpha = 0.05$ is 3.47, a little higher than the 3.35 for 30 plots because the error term has fewer degrees of freedom. Our F of 14.60 is more than four times that cutoff. Values this large arise by chance less than 0.1% of the time, so we reject the null hypothesis that all three group means are equal.

Effect size: eta-squared

The F-test answers whether the group means differ by more than chance would produce. It says nothing about *how much* of the variation in salamander abundance forest type explains. Chapter 5 made the same distinction, where p told us whether a difference was detectable and Cohen's d told us how large it was. Chapter 5 measured effect size in the study's units or in standard deviations. ANOVA adds a third kind, the share of the total variation that the groups explain.

ANOVA's effect size is **eta-squared** (η^2), and we already computed both of its ingredients when we partitioned the sums of squares:

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}} = \frac{SS_{\text{Between}}}{SS_{\text{Between}} + SS_{\text{Within}}}$$

SS_{Between} is the variation the group means explain and SS_{Total} is all the variation there is. So η^2 is the proportion of the total variation in the response that the factor explains. It runs from 0 to 1.

Table 11: Eta-squared for the effect of forest type on salamander abundance.

SS Between	SS Within	SS Total	Eta-squared
312.58	224.75	537.33	0.582

Here $\eta^2 = 0.58$, so forest type explains about 58% of the variation in salamander abundance across these 24 plots. The remaining 42% is within-group variation, the plot-to-plot differences that forest type alone cannot explain.

η^2 is the same quantity as R^2 in regression (Chapter 9), computed the same way. Both divide the total variation into a part the model explains and a part it leaves over, and report the explained part as a proportion of the total.

Interpreting the number. Cohen's conventional benchmarks are $\eta^2 = 0.01$ (small), 0.06 (medium), and 0.14 (large), so 0.58 is a large effect by these benchmarks. Treat those cutoffs as rough orientation. What counts as a large effect depends on the system you are studying. An η^2 of 0.10 would be unremarkable in a controlled greenhouse experiment and notable in a field survey where dozens of uncontrolled factors also drive the response.

p depends on sample size, and η^2 does not. Survey enough plots and a difference of half a salamander per plot will clear $p < 0.05$ while explaining almost none of the variation. Reporting both keeps a statistically detectable result from being mistaken for an ecologically meaningful one.

Post-hoc comparisons with Tukey HSD

The omnibus F test tells us that at least one group mean differs, but not which pairs. **Post-hoc comparisons** are tests run after a significant omnibus test to find which pairs of groups differ. This is a different use of "post-hoc" from Chapter 5's post-hoc power. To identify which forest types differ, we use **Tukey's Honestly Significant Difference (HSD)** test (Tukey 1949). Tukey HSD compares every pair of groups while keeping the familywise error rate across all of the comparisons at α, so it is the standard choice after a significant ANOVA.

```
TukeyHSD(aov_out)
```

Table 12: Tukey HSD post-hoc comparisons for salamander abundance by forest type.

Comparison	diff	lwr	upr	p adj
Secondary-Old-growth	-6.88	-11.00	-2.75	0.001
Managed-Old-growth	-8.25	-12.37	-4.13	< 0.001
Managed-Secondary	-1.37	-5.50	2.75	0.683

Reading the table, column by column. Each row is one pairwise comparison, and there are three of them because three groups make three pairs.

- `diff` is the difference in group means, computed as *the first group in the row label minus the second*: `Secondary-Old-growth` is secondary minus old-growth.
- `lwr` and `upr` are the lower and upper bounds of the 95% confidence interval for that difference.
- `p adj` is the p-value *after* adjusting for the fact that you are making three comparisons at once. Because of this adjustment, the chance of at least one false positive across all three comparisons stays at 5%, compared with the 14% we computed at the start of the chapter for three unadjusted tests.

Watch the sign. The row `Secondary-Old-growth` has a diff of -6.88, and that negative sign does *not* mean secondary forest has more salamanders. It means secondary *minus* old-growth is negative, so old-growth is the higher of the two, by 6.88 individuals per plot. Always translate the sign back into a sentence naming which group is larger before you write anything down.

The confidence interval and the p-value say the same thing. A pair differs significantly at $\alpha = 0.05$ exactly when its 95% interval excludes zero. Check both columns against each other as a habit: `Secondary-Old-growth` runs from -11.00 to -2.75, entirely below zero, and its p is 0.001. `Managed-Secondary` runs from -5.50 to 2.75, straddling zero, and its p is 0.683. The two columns always agree, so if they seem not to, reread that row.

So the three comparisons give:

- **Old-growth vs. Secondary.** Old-growth is higher by 6.88 individuals/plot, $p = 0.001$, a significant difference.
- **Old-growth vs. Managed.** Old-growth is higher by 8.25 individuals/plot, $p < 0.001$, a significant difference.
- **Secondary vs. Managed.** They differ by 1.37 individuals/plot, $p = 0.683$, not a significant difference.

A non-significant difference is one the test could not detect. Secondary and managed plantation differ by 1.37 salamanders per plot in this sample. That difference is present in the sample. The test says that with eight plots per group, we cannot distinguish it from sampling noise. A larger study might well separate them. Report it as "we did not detect a difference," never as "there is no difference."

Compact letter display. A compact letter display summarizes post-hoc results with a letter above each group. With more than three groups, a list of pairwise sentences gets long, so this is the usual summary. Groups that share a letter were not significantly different, and groups that share no letter were. Figure 8 puts the salamander results in that form.

Old-growth gets its own letter because it differs from both of the others. Secondary and managed share a letter because the test could not separate them. Sometimes a group carries *two* letters, written "ab". Suppose group 1 is labeled "a" and group 3 is labeled "b" because those two differ significantly. Group 2 has a mean between them. If group 2 does not differ significantly from group 1 or from group 3, it shares a letter with each, so it is labeled "ab".

In these invented counts, old-growth plots had the highest salamander abundance. Secondary forest and managed plantation could not be told apart with 8 plots of each type, even though we built in a difference of about 2 salamanders per plot between them.

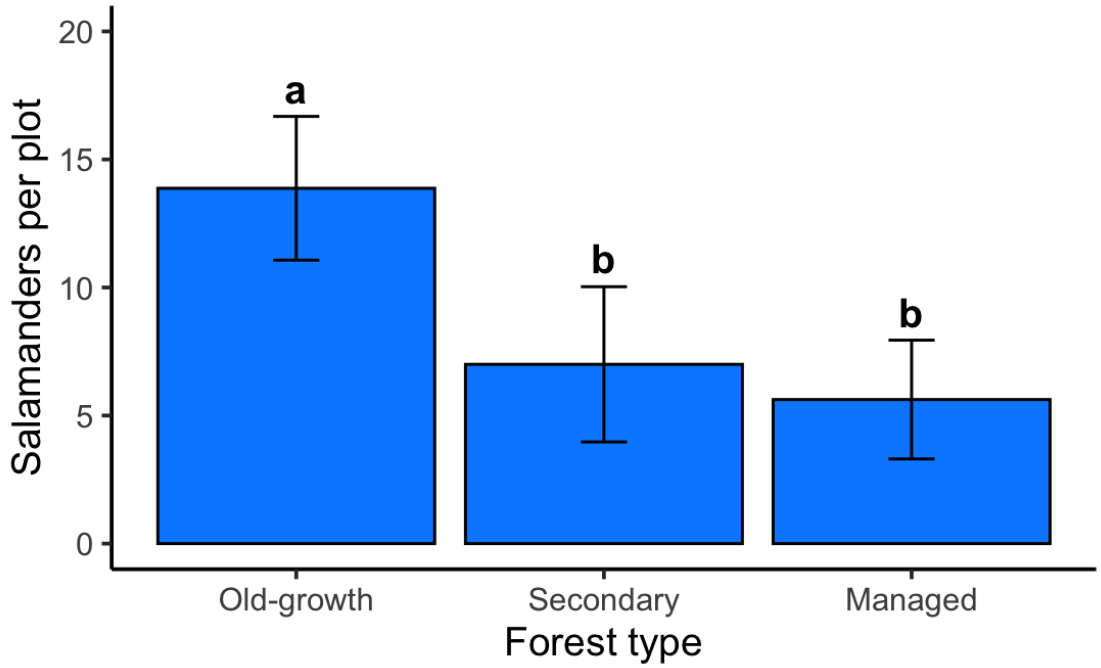


Figure 8: Mean salamander abundance with Tukey letters. Old-growth carries an ‘a’ and differs from both other types. Secondary and managed share a ‘b’, so they were not distinguishable from each other. Error bars show 95% confidence intervals.

! Results reporting: one-way ANOVA

Report the direction in plain language, the test statistic with both degrees of freedom, p, and the effect size, then which groups the post-hoc comparisons separated.

> Salamander abundance differed significantly among forest types, $F(2, 21) = 14.60$, $p < 0.001$, $\eta^2 = 0.58$. Tukey HSD comparisons showed more salamanders in old-growth plots than in secondary forest, $p = 0.001$, and managed plantation, $p < 0.001$. Secondary forest and managed plantation did not differ significantly, $p = 0.683$.

Table 13 collects the one-way ANOVA on one card.

Table 13: Test card: one-way ANOVA.

Purpose	Do the means of two or more separate groups differ? One factor, different units in each group
Hypotheses	$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$, H_a: at least one mean differs
Assumptions	Approximately normal residuals, and similar spread in every group (homogeneity of variance)
Test statistic	$F = MS_{\text{Between}}/MS_{\text{Within}}$, $df_{\text{Between}} = k - 1$, $df_{\text{Within}} = N - k$
Report	$F(df_{\text{Between}}, df_{\text{Within}})$, p, and $\eta^2 = SS_{\text{Between}}/SS_{\text{Total}}$. After a significant F, Tukey HSD comparisons naming which groups differ

Nonparametric alternative: the Kruskal-Wallis test

Our assumption checks above came back clean, so the ANOVA result stands as-is. Suppose instead the groups had visibly unequal spread. Then Welch's ANOVA (`oneway.test()` in R) compares the means without assuming equal variances, the same fix Welch's t-test makes in Chapter 6. If the residuals were clearly skewed in a small or unbalanced sample, the **Kruskal-Wallis test** (Kruskal and Wallis 1952) is the rank-based, nonparametric alternative to one-way ANOVA. It has the same relationship to ANOVA that the Mann-Whitney-Wilcoxon test has to the independent-samples t-test. Kruskal-Wallis uses the Mann-Whitney-Wilcoxon logic, extended to more than two groups.

How it works, conceptually. Rank every observation across all groups together, from smallest to largest, ignoring group membership. Then check whether the average rank differs more across groups than chance alone would produce. If forest type truly matters, one group's observations should occupy systematically higher or lower ranks than the others. In the salamander data, the old-growth plots have a mean rank of 19.7 out of 24, against 10.1 for secondary and 7.8 for managed. If forest type had no effect, all three mean ranks would be close to 12.5, the middle of 1 to 24.

Its assumption. Kruskal-Wallis assumes the groups' distributions have roughly the same shape, and it does not require normality. When the shapes match, a significant result means the groups differ in location, meaning where the center of each distribution is, measured by the median, as with the Mann-Whitney-Wilcoxon test. If the shapes differ a lot, a significant result tells you the groups differ, but not whether that difference is in location, spread, or shape.

```
kruskal.test(abundance ~ forest_type, data = salamander_df)
```

Table 14: Kruskal-Wallis test on salamander abundance by forest type.

Statistic (χ^2)	df	p
13.09	2	0.001

$\chi^2(2) = 13.09$, $p = 0.001$, a significant result and the same conclusion as the ANOVA ($F(2, 21) = 14.60$, $p < 0.001$). The two tests agree. That is expected when the parametric assumptions hold, and ours did here.

Post-hoc comparisons. Just as a significant ANOVA needs Tukey HSD to identify *which* groups differ, a significant Kruskal-Wallis result needs its own follow-up, pairwise rank-sum tests (Mann-Whitney-Wilcoxon, applied to each pair) with a correction for running multiple comparisons (Holm 1979).

```
pairwise.wilcox.test(salamander_df$abundance, salamander_df$forest_type,
                     p.adjust.method = "holm")
```

Table 15: Pairwise Mann-Whitney-Wilcoxon comparisons (Holm-adjusted).

Group 1	Group 2	p
Secondary	Old-growth	0.011
Managed	Old-growth	0.004
Managed	Secondary	0.486

The pattern matches Tukey HSD exactly. Old-growth differs significantly from both Secondary ($p = 0.011$) and Managed ($p = 0.004$), while Secondary and Managed do not differ from each other ($p = 0.486$). The two

approaches use different assumptions and different calculations, and they agree here because the parametric assumptions held.

Table 16 collects the Kruskal-Wallis test on one card.

Table 16: Test card: Kruskal-Wallis test.

Purpose	Do two or more separate groups differ? Skewed residuals in a small or unbalanced sample
Hypotheses	H_0: group has no effect, so every group's mean rank is close to the middle rank. H_a: at least one group ranks systematically higher or lower
Assumptions	Normality not required. Reads as a difference in location (median) only when the groups' distributions have similar shape
Test statistic	Rank all observations together and compare the mean ranks across groups. R reports the statistic as χ^2, $df = k - 1$
Report	$\chi^2(df)$, p. After a significant result, pairwise Mann-Whitney-Wilcoxon tests with a Holm correction

Repeated-measures ANOVA

Every unit in the salamander study was measured once: 24 plots, 8 per forest type. Many environmental studies instead measure the same monitoring stations in every season or the same streams before, during, and after a drought. A **repeated-measures** (within-subjects) design measures the same units under every condition, and it calls for a repeated-measures ANOVA.

This is ANOVA's version of the paired t-test from Chapter 6. The paired t-test subtracted each plot's before score from its own after score, so differences between plots dropped out and only the change was left. With three or more conditions there is no single difference to take, so repeated-measures ANOVA does the equivalent with sums of squares, measuring how much the units differ from one another and removing that variation from the error term.

Fine particulate matter at four monitoring stations

An air quality agency has four PM2.5 monitors, ordered from the most rural (A) to the most urban (D). The concentrations in this example are invented. Table 17 gives each monitor's mean PM2.5 concentration in winter, spring, and summer. The question is whether concentrations differ by season: $H_0 : \mu_{\text{Winter}} = \mu_{\text{Spring}} = \mu_{\text{Summer}}$ vs. H_a: at least one seasonal mean differs.

Table 17: Seasonal mean PM2.5 (µg/m³) at four monitoring stations, with each station's mean and each season's mean.

Station	Winter	Spring	Summer	Station mean
A	9	2	4	5
B	9	8	7	8
C	16	11	12	13
D	18	11	13	14
Season mean	13	8	9	10

Figure 9 draws the same numbers, one line per station. Winter is the highest season at every station. But the stations themselves differ far more than the seasons do. Station A averages 5 µg/m³ and station D averages 14.

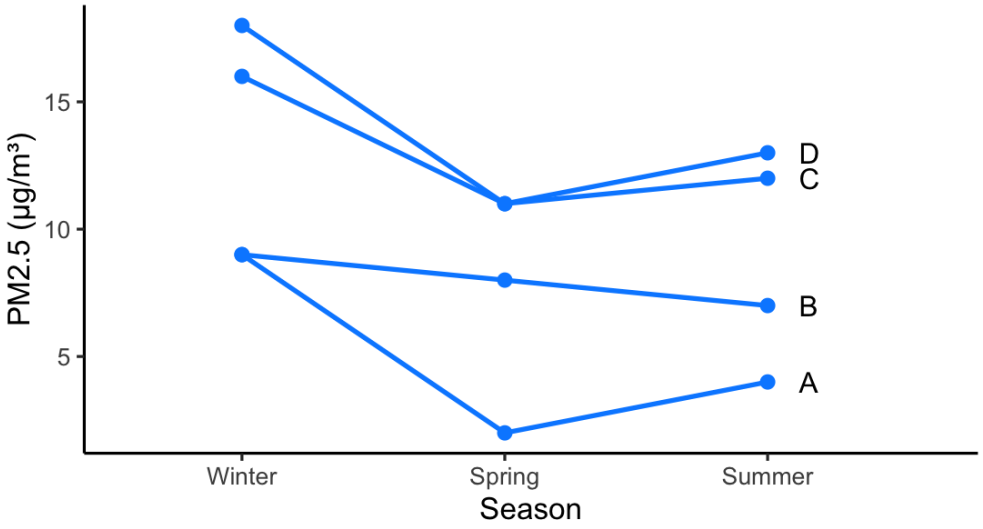


Figure 9: Seasonal mean PM2.5 at four stations, one line per station. Each line is highest in winter, and the lines are far apart.

Splitting off the stations

A one-way ANOVA on these 12 numbers would split the variation into two pieces, as before: $SS_{\text{Total}} = 230$, $SS_{\text{Between}} = 56$ for seasons, and $SS_{\text{Within}} = 174$ left over. It would also break the independence assumption, because each station contributes three of the 12 numbers, and readings from the same station tend to be alike.

Repeated-measures ANOVA splits SS_{Within} one more time. Part of it is station-to-station variation. The units measured repeatedly are called subjects, even when they are stations, so this piece is SS_{Subjects}, computed exactly like SS_{Between} but from the station means instead of the season means. Each station mean counts once per season, so

$$SS_{\text{Subjects}} = 3\left[(5-10)^2 + (8-10)^2 + (13-10)^2 + (14-10)^2\right] = 3(54) = 162.$$

Whatever is left is the error term:

$$SS_{\text{Error}} = SS_{\text{Within}} - SS_{\text{Subjects}} = 174 - 162 = 12,$$

so the full split is

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Subjects}} + SS_{\text{Error}}, \qquad 230 = 56 + 162 + 12.$$

SS_{Error} is the part of the within-season variation that the stations' own levels can't explain. It measures how much each station's seasonal pattern departs from the average seasonal pattern. Its degrees of freedom are $(k-1)(n-1) = (3-1)(4-1) = 6$, where n is the number of stations. The stations use up 3 of the 9 degrees of freedom the one-way ANOVA would have had.

F uses the same numerator as before, $MS_{\text{Between}} = 56/2 = 28$. Only the denominator changes:

$$\text{One-way: } F = \frac{28}{174/9} = \frac{28}{19.33} = 1.45 \qquad \text{Repeated measures: } F = \frac{28}{12/6} = \frac{28}{2} = 14.00$$

The repeated-measures test finds a significant seasonal effect, $F(2, 6) = 14.00$, $p = 0.005$. The one-way ANOVA would give $F(2, 9) = 1.45$, $p = 0.285$, and miss it. Removing the stations shrank the error mean square from 19.33 to 2, and F grew by the same factor. A repeated-measures design has more power than a between-subjects design with the same number of measurements when the units differ from one another, as monitoring stations usually do. The power comes from removing the unit-to-unit differences from the error term, so they are no longer counted as noise.

Chapter Summary

Why it matters. Comparing more than two groups with a stack of t-tests inflates your false-positive rate fast. ANOVA solves that by asking one omnibus question first, and its variance-partitioning logic is the same idea that reappears in regression, factorial designs, and ANCOVA.

Core ideas

- **Many t-tests inflate the false-positive rate.** Each comparison carries its own chance of a false positive, and they accumulate. With m comparisons the familywise error rate is $1 - (1 - \alpha)^m$, which is already 14% for three groups.
- **Partition the variation.** Total variation splits into a piece explained by group membership (SS_{Between}) and unexplained residual variation (SS_{Within}).
- **F is a ratio of the two.** After dividing each sum of squares by its degrees of freedom to get a mean square, $F = MS_{\text{Between}}/MS_{\text{Within}}$. A large F means between-group differences are large relative to within-group noise.
- **Run the omnibus test first, then post-hoc comparisons.** The omnibus test asks whether *any* group means differ. If it is significant, run post-hoc comparisons (Tukey HSD) to find out *which* pairs differ.
- **η^2 is the effect size.** $SS_{\text{Between}}/SS_{\text{Total}}$ is the proportion of variation explained by the factor. It is the same quantity as regression's R^2, and it does not depend on sample size the way p does, so report it alongside F.
- **Check the assumptions.** ANOVA needs approximately equal variances across groups (Levene's test) and roughly normal residuals. Treat formal tests as evidence rather than a gate, since they have least power exactly when small samples make violations matter most.
- **When assumptions fail.** Welch's ANOVA handles unequal spread. The Kruskal-Wallis test is the rank-based alternative for skewed residuals in small samples, with `pairwise.wilcox.test()` as its post-hoc counterpart.
- **Repeated measures.** When the same units are measured under every condition, repeated-measures ANOVA splits SS_{Within} into SS_{Subjects} and a smaller SS_{Error}. The smaller error term is why it has more power, the same reason the paired t-test does.

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